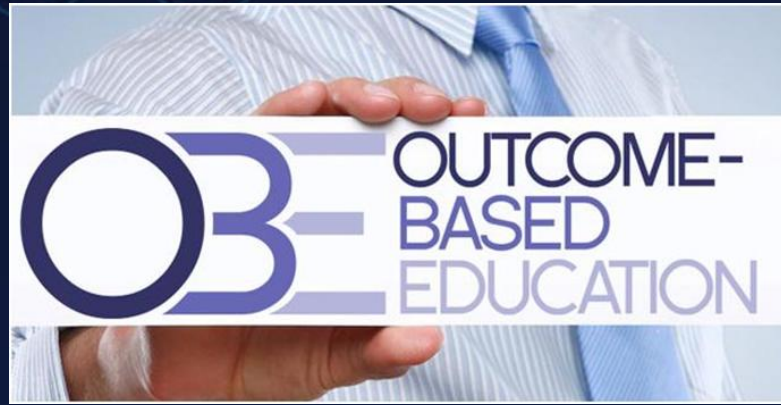


# CSE 3200: System Development Project

## CLO–Assessment Mapping & Rubric-Based Evaluation



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# Learning Objectives

By the end of this presentation, participants will be able to:

1. **Analyze** complex engineering problems and identify system requirements.
2. **Design and develop** a complete software/system solution using appropriate methodologies
3. **Apply modern tools and technologies** in system implementation
4. **Test, validate, and evaluate** system performance and reliability
5. **Work effectively in teams** and manage project tasks professionally
6. **Prepare technical documentation** and deliver effective presentations





# Course Overview

- ❑ Project-based course (individual/group)
- ❑ Focus on solving **complex engineering problems**
- ❑ Includes:
  - ✓ Computing system/software development
  - ✓ OS modules / I/O drivers / real-world applications
- ❑ Emphasis on:
  - ✓ Software/Hardware engineering principles
  - ✓ Documentation & presentation

# Course Learning Outcomes (CLOs)

After completing the course, students will be able to:

- ✓ **CLO1:** Problem analysis
- ✓ **CLO2:** System design & development
- ✓ **CLO3:** Use of modern tools
- ✓ **CLO4:** Testing & investigation
- ✓ **CLO5:** Teamwork & project management
- ✓ **CLO6:** Communication & documentation

# Assessment Components

| Assessment                         | Marks      |
|------------------------------------|------------|
| ✓ Proposal (A <sub>1</sub> )       | 10         |
| ✓ Design (A <sub>2</sub> )         | 15         |
| ✓ Implementation (A <sub>3</sub> ) | 20         |
| ✓ Testing (A <sub>4</sub> )        | 15         |
| ✓ Documentation (A <sub>5</sub> )  | 15         |
| ✓ Presentation (A <sub>6</sub> )   | 15         |
| ✓ Teamwork (A <sub>7</sub> )       | 10         |
| <b>Total</b>                       | <b>100</b> |



# CLO–Assessment Mapping

- ✓ CLO1 → A1
- ✓ CLO2 → A2 + A3
- ✓ CLO3 → A3
- ✓ CLO4 → A4
- ✓ CLO5 → A2 + A7
- ✓ CLO6 → A5 + A6

# Evaluation Methodology

1. Use **rubrics for each CLO**
2. Assign marks per assessment
3. Map marks → CLOs
4. Convert to percentage
5. Determine attainment level

# Performance Levels

| Level | Description  | Range       |
|-------|--------------|-------------|
| 4     | Excellent    | $\geq 80\%$ |
| 3     | Good         | 65–79%      |
| 2     | Satisfactory | 50–64%      |
| 1     | Poor         | $< 50\%$    |



# CLO1 Rubric (Problem Analysis)

- ☐ Problem definition
- ☐ Requirement analysis
- ☐ Feasibility study
- ✓ Clear & complete → Level 4
- ✓ Partial → Level 2–3
- ✓ Poor → Level 1

| Criteria             | 4 (Excellent)               | 3 (Good)        | 2 (Satisfactory)           | 1 (Poor)   |
|----------------------|-----------------------------|-----------------|----------------------------|------------|
| Problem Definition   | Clear, precise, well-scoped | Mostly clear    | Vague or partially defined | Unclear    |
| Requirement Analysis | Complete & justified        | Mostly complete | Some missing               | Incomplete |
| Feasibility Study    | Comprehensive analysis      | Moderate        | Limited                    | Not done   |

# CLO2 Rubric (Design & Development)

- ☐ System architecture
- ☐ Algorithm/logic
- ☐ Design diagrams (UML)
- ✓ Scalable & optimized → Level 4
- ✓ Basic → Level 2

# CLO<sub>3</sub> Rubric (Modern Tools)

- ☐ Tool selection
- ☐ Tool usage
- ☐ Integration
- ✓ Advanced usage → Level 4
- ✓ Limited usage → Level 2



# CLO<sub>4</sub> Rubric (Testing)

- ☐ Test cases
- ☐ Validation
- ☐ Performance analysis
- ✓ Comprehensive → Level 4
- ✓ Minimal → Level 2

# CLO5 Rubric (Teamwork)

☐ Collaboration

☐ Task management

☐ Leadership

✓ Strong teamwork → Level 4

✓ Weak participation → Level 1

# CLO6 Rubric (Communication)

- ☐ Report quality
- ☐ Technical content
- ☐ Presentation
- ☐ Q&A
- ✓ Professional → Level 4
- ✓ Poor → Level 1



# Example Project

Project: Smart Traffic Monitoring System

Student: A

# Sample Marks

| Assessment | Marks |
|------------|-------|
| A1         | 8/10  |
| A2         | 12/15 |
| A3         | 16/20 |
| A4         | 11/15 |
| A5         | 13/15 |
| A6         | 12/15 |
| A7         | 9/10  |

# Justification of Sample Marks

## Basis of Evaluation (Rubric-Based Assessment)

- ✓ Marks are assigned based on **predefined rubrics for each CLO**
- ✓ Each assessment evaluates **specific performance criteria**
- ✓ Scores reflect **quality, completeness, and effectiveness**



# Assessment-wise Justification

## ❑ A1: Proposal (8/10)

- ✓ Problem well-defined and relevant
- ✓ Minor gaps in feasibility analysis

## ❑ A2: Design (12/15)

- ✓ Good system architecture and UML diagrams
- ✓ Lacks optimization and scalability details

## ❑ A3: Implementation (16/20)

- ✓ Functional and mostly efficient code
- ✓ Limited advanced features / optimization

## ❑ A4: Testing (11/15)

- ✓ Basic test cases and validation present
- ✓ Incomplete performance analysis

## ❑ A5: Documentation (13/15)

- ✓ Well-structured report with clear explanation
- ✓ Minor formatting and detail issues

## ❑ A6: Presentation & Viva (12/15)

- ✓ Clear presentation and good demonstration
- ✓ Moderate performance in Q&A session

## ❑ A7: Teamwork (9/10)

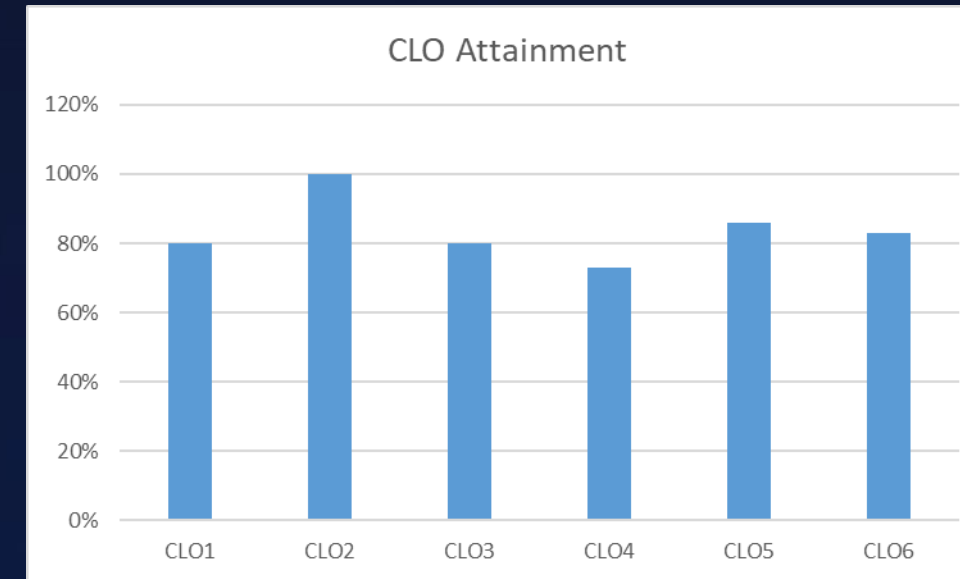
- ✓ Strong collaboration and task management
- ✓ Minor imbalance in contribution

# CLO Calculation Example

- ✓ CLO1 = 08/10 → 80%
- ✓ CLO2 = 20/20 → 100%
- ✓ CLO3 = 08/10 → 80%
- ✓ CLO4 = 11/15 → 73%
- ✓ CLO5 = 13/15 → 86%
- ✓ CLO6 = 25/30 → 83%

# CLO Attainment Summary

| CLO  | %    | Level |
|------|------|-------|
| CLO1 | 80%  | 4     |
| CLO2 | 100% | 4     |
| CLO3 | 80%  | 4     |
| CLO4 | 73%  | 3     |
| CLO5 | 86%  | 4     |
| CLO6 | 83%  | 4     |





# Interpretation



☒ All CLOs attained ( $\geq 60\%$ ) ✓

☐ Strong in:

- ✓ Design
- ✓ Communication

☐ Improvement needed:

- Testing

# Class Attainment (Example)

○ CLO1: 87% students attained

○ CLO2: 93%

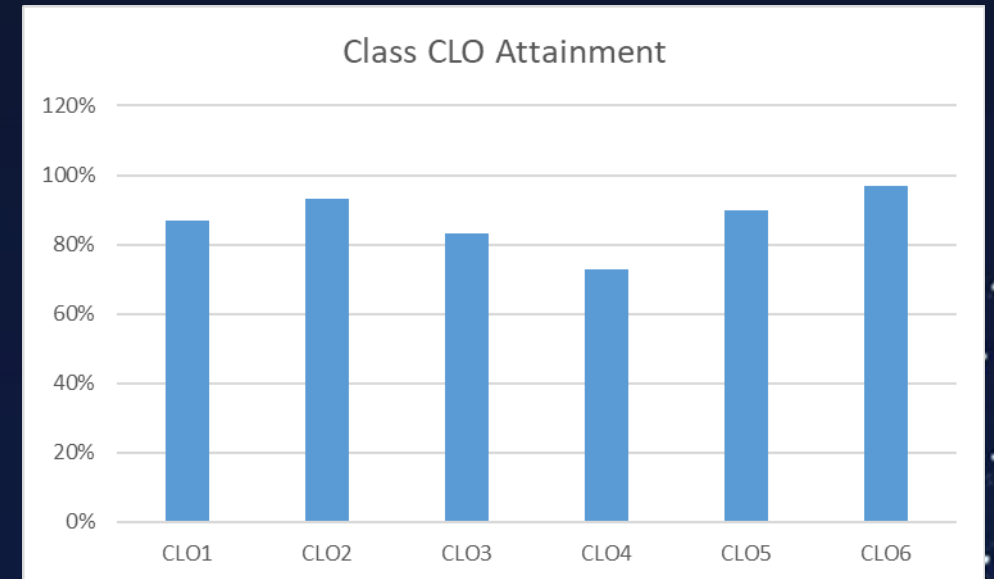
○ CLO3: 83%

○ CLO4: 73%

○ CLO5: 90%

○ CLO6: 97%

✓ All CLOs achieved



# Summary

## ❑ Rubric-based evaluation ensures:

- ✓ Transparency
- ✓ Fairness
- ✓ Measurability

## ❑ CLO attainment:

- ✓ Supports **OBE & BAETE accreditation**
- ✓ Enables continuous improvement





# Thank you for your Attention

